Topic 6

Multirate Digital Signal Processing 2
Decimation 2
Interpolation 7
Linear Interpolation 11
Sampling rate conversion by Non-integer factors 12
Illustration of Interpolation by a factor 3/2 14
Decimation by 2 15
Interpolation by 2 16

Multirate Digital Signal Processing

The increasing need in modern digital systems to process data at more than one sampling rate has led the development of a new sub-area in DSP known as multirate signal processing. The two primary operations in multirate proceeding are

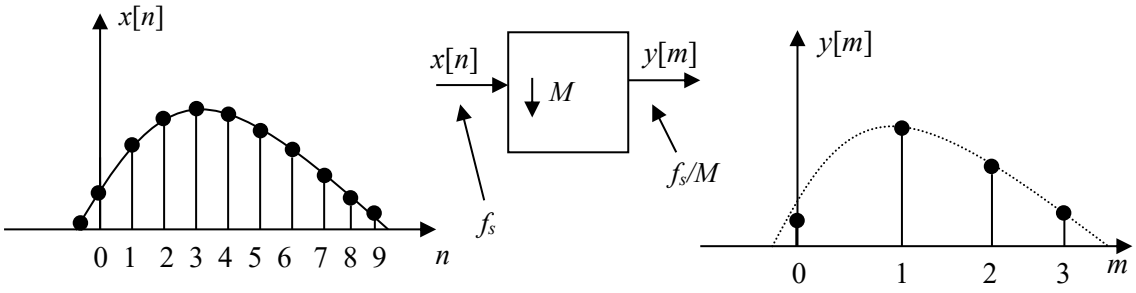
- **Decimation**
  (We can decrease the sampling rate f_s of a given signal $x[n]$: This is sometimes called **down sampling**).

- **Interpolation**
  (We can increase the sampling rate f_s of a given signal $x[n]$: This is sometimes called up **sampling**).

The process of decimation and interpolation are the fundamental operations in multirate signal processing.

Decimation

A sampling rate decreaser is shown below. We should confine our attention to a decrease by an integer factor M. (e.g. $M = 3$)



A discrete-time signal, x[n], is down-sampled by a factor of M=3, yielding another discrete-time signal, y[n].

Example: $x[n] = \{1, 2, 4, 3, 5, -6, -8, 2, -3, 2\}$

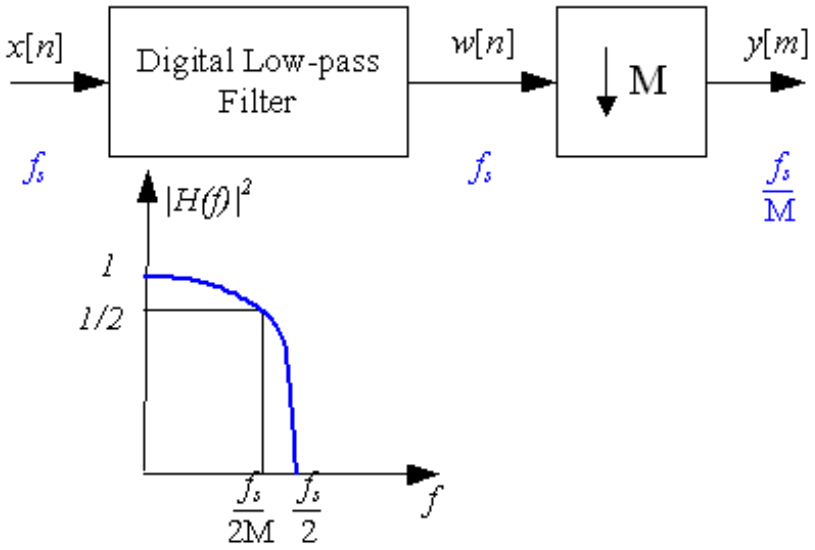
$\downarrow$ Down sample by 2

$y[m] = \{1, 4, 5, -8, -3\}$

The output signal $y[m]$ is obtained by taking every M^{th} sample of the input signal. If $M = 3$, we should just take every third sample of $x[n]$ to form the desired signal $y[m]$.

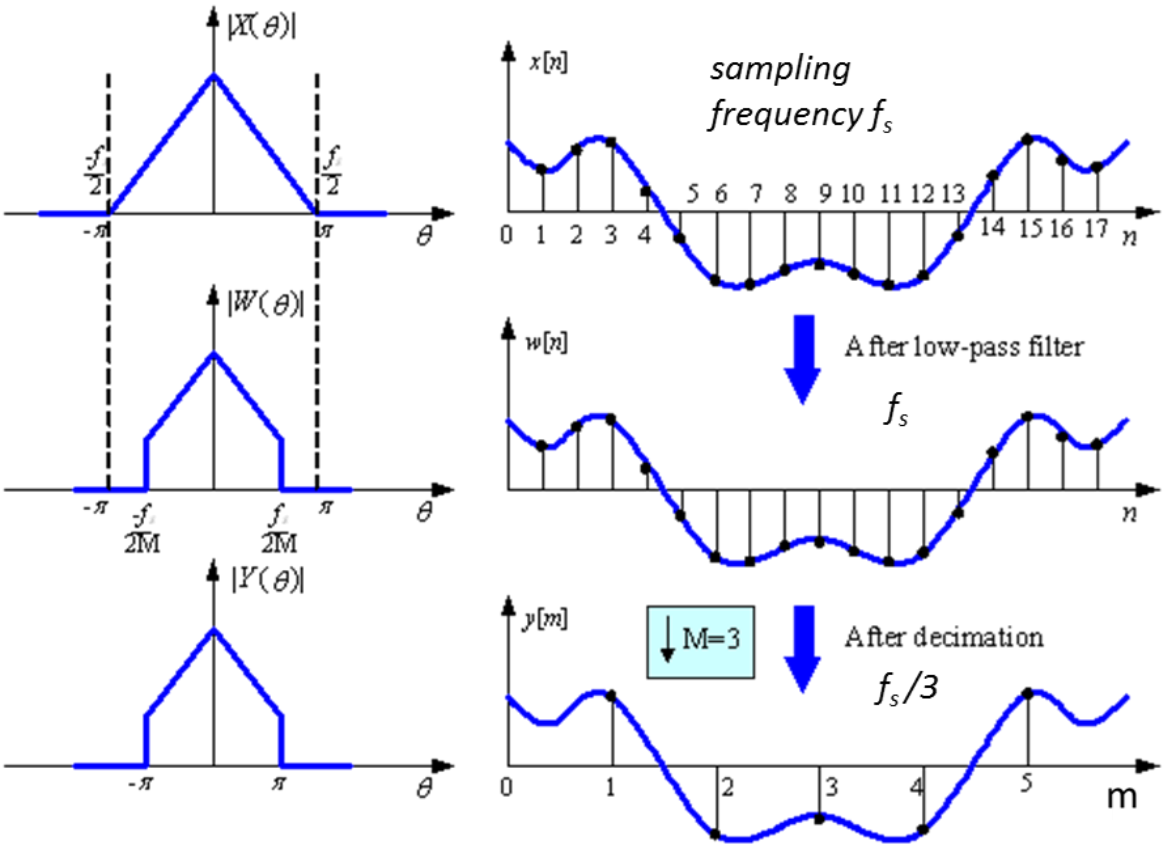
Obviously, it only makes sense to reduce the sampling rate if the information constant of the signal we wish to preserve is band limited to $\frac{f_s}{6}$;

Figure below shows a block representation of a times M decimator. We see that the signal $x[n]$ is first passed through a low pass filter that attenuates the band from $\frac{f_s/2}{M} = \frac{f_s}{2M}$ to $\frac{f_s}{2}$ to prevent aliasing

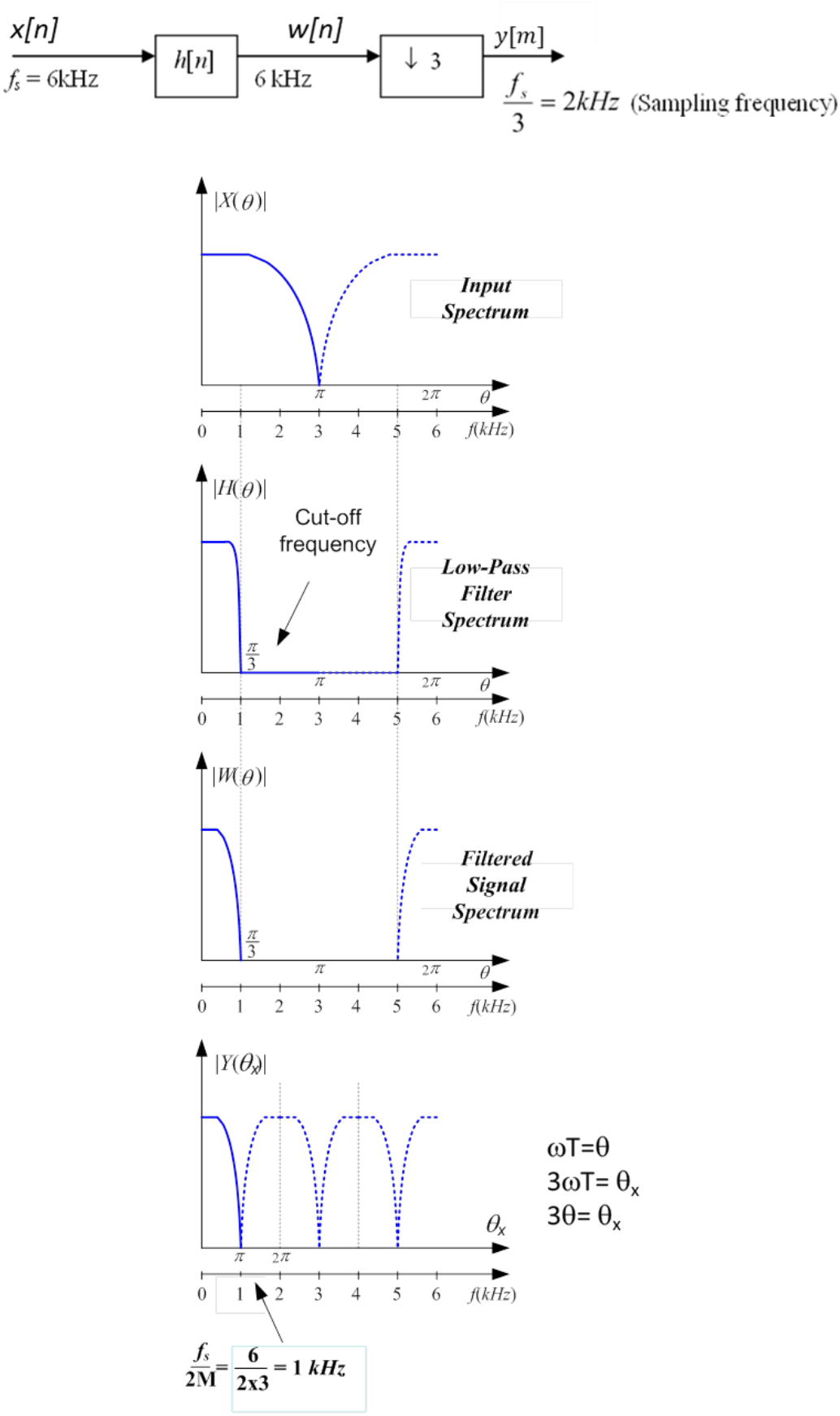


Prior to down-sampling, a discrete-time signal must first go through a low-pass filter to remove its high-frequency components.
This prevents aliasing.

The spectral description of the decimation process is shown below



Spectral description of decimation process



Spectral interpretation of decimation of a signal from 6kHz to 2kHz

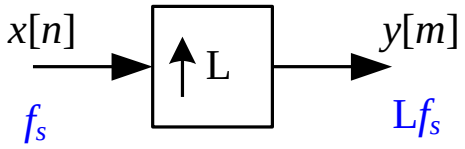
Note: Such a processing requirement may arise, for example when a speech signal is over sampled at f_s = 32 kHz. Since we are interested only in a band of 0-4 kHz, sampling rate can be reduced to 8 kHz, so the first step in the decimation process has to be the digital filtering of the signal $x[n]$ is band limited to $f_s/8$. Should we use IIR (recursive) or FIR (non-recursive) for the low pass filtering required?

Using an IIR filter in this case has an obvious shortcoming. We cannot take advantage of the fact that we only have to compute every N^{th} output, since previous outputs are required to compute the Mth output. Thus no saving realized. On the other hand, *using an FIR filter*, in this case implies that we can do our computations at the rate $\frac{f_s}{M}$. Thus, using an FIR filter in the decimation process will lead to a significantly lower computation rate.

Another advantage of using an FIR filter is the fact that we can easily design linear phase filters and this is desirable in many applications.

Interpolation

The process of interpolation involves a sampling rate increase (see diagram below)



A general block diagram representation of up-sampling process.

Example:

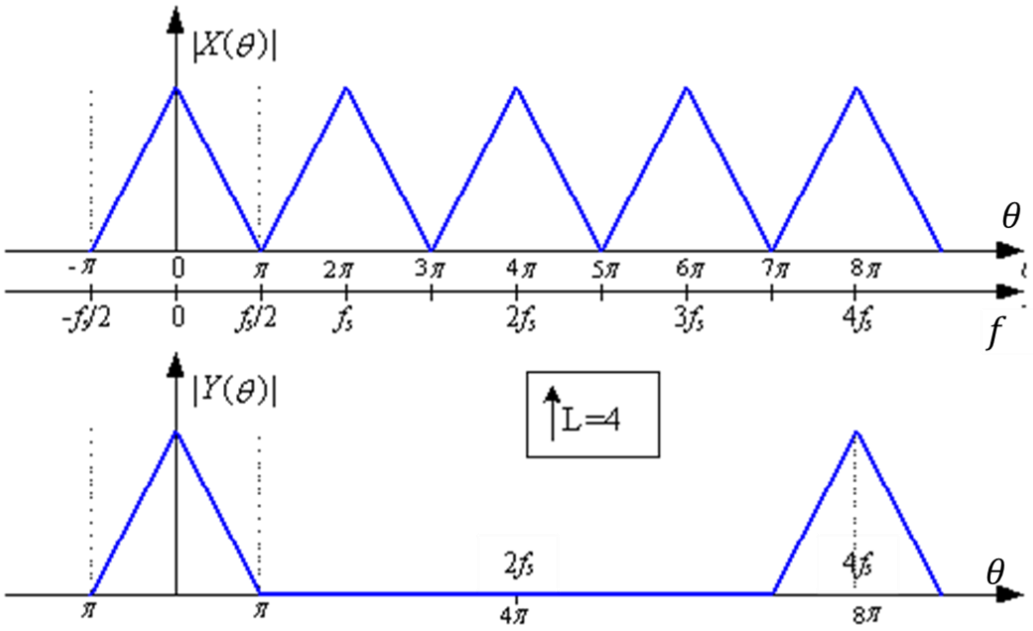
$x[n]=\{1,2,4,3,-5,6,-7,2,4,3\}$

$\uparrow 2$

$y[m]=\{1,0,2,0,4,0,3,0,-5,0,6,0,-7,0,2,0,4,0,3,0\}$

The sequence $x[n]$ was derived by sampling $x(t)$ at a sampling rate of f_s and we want to obtain a sequence $y[m]$ that approximates as closely as possible the sequence that would have been obtained had we sampled $x(t)$ at the rate Lf_s.

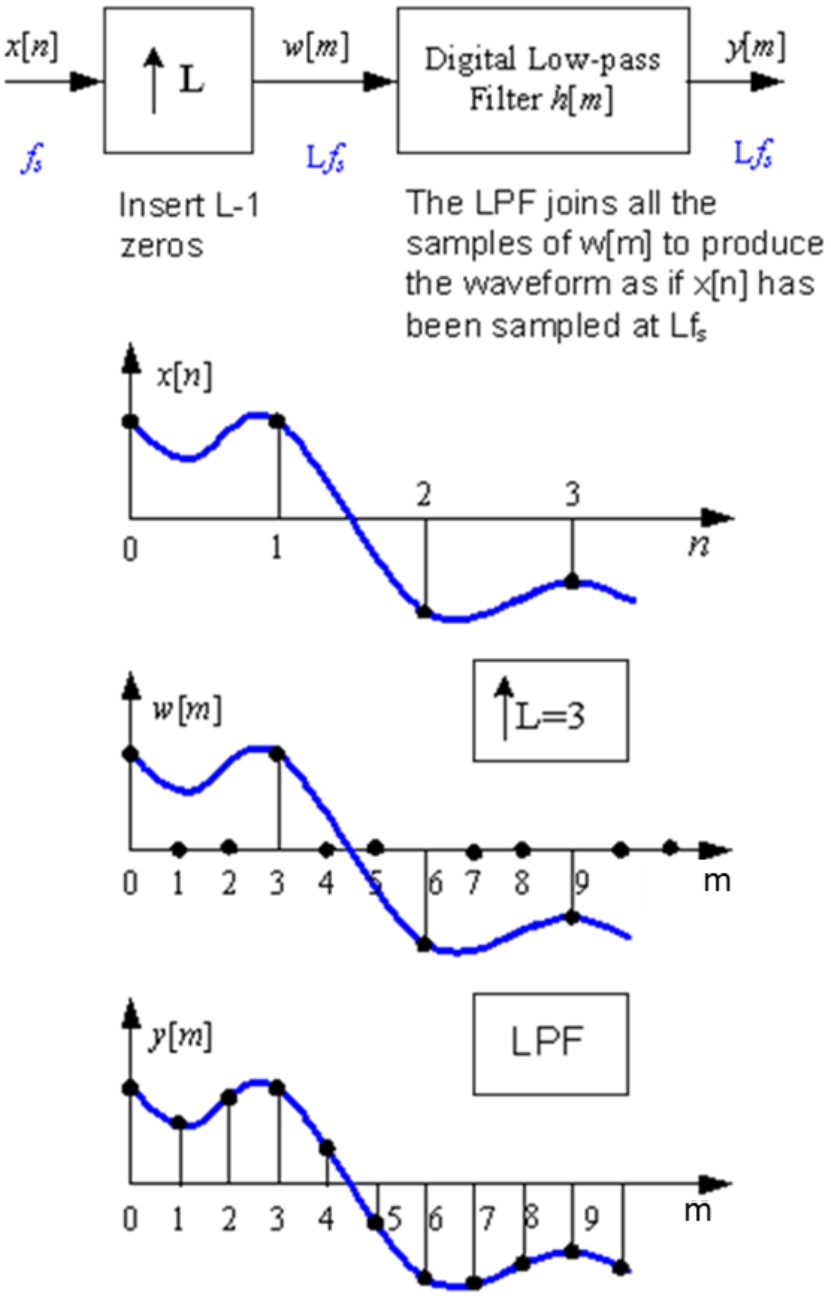
This involves inserting between any two samples $x[n]$ and $x[n\text{-}1]$ an additional L-1 samples.



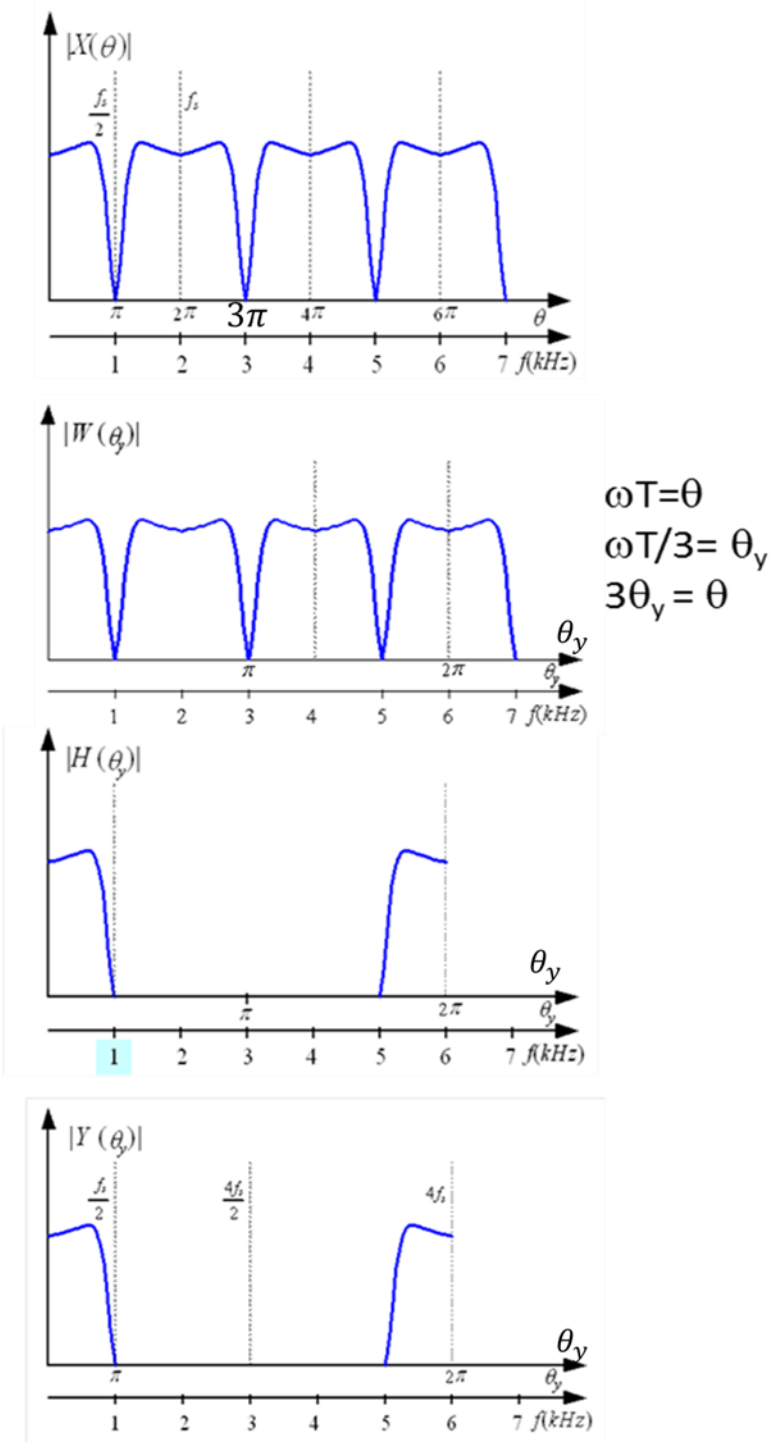
Spectral description of the interpolation process. Sampling frequency of y(n) = 4f_s .

We observe from above Figure that to go from $X(\theta)$ to $Y(\theta)$, we have to pass $x[n]$ through a lowpass digital filter designated at the Lf_s sampling rate that attenuates sufficiently any frequency components above $\frac{f_s}{2}$.

The frequency domain interpretation of the process is given below.



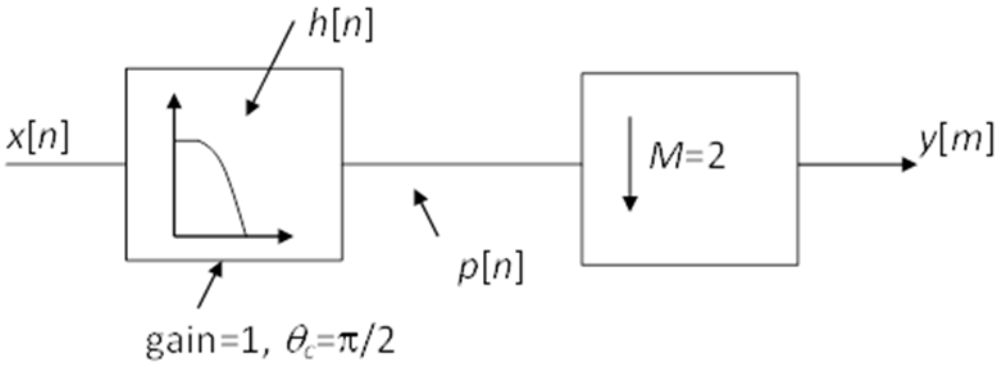
A general description of the interpolation process. A discrete-time signal is up-sampled by inserting zeros and then filtered by a low-pass filter.
Note that for each sample of x[n], three output samples y[m] are obtained.



Spectral interpretation of interpolation of a signal from 2kHz to 6 kHz.

Note: Obviously the same reasoning that led us to believe that FIR filters are preferable in the decimation holds here also.

Example: Decimation, $x[n] = \{2, 6, 4, 2, 6, 8, 4, 2, 4, 4\}$

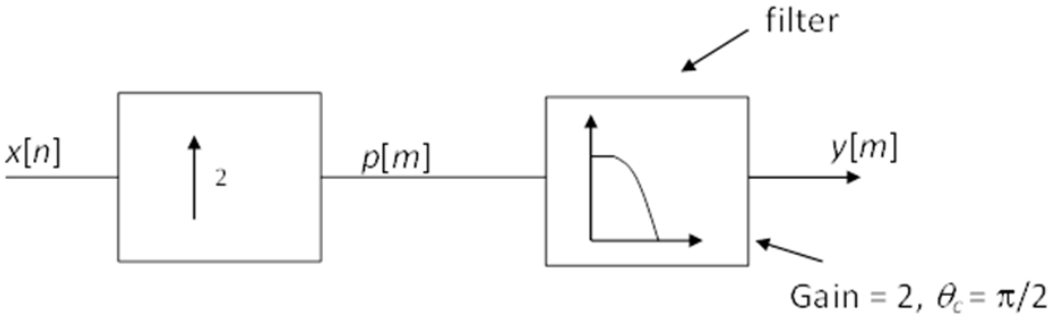


$$h[n] = \left\{\frac{1}{2}, \frac{1}{2}\right\}$$

$$p[n] = x[n] * h[n] = \{1,4,5,3,4,7,6,3,3,4,2\}$$

When p[n] is downsampled by 2, we obtain $y[n] = \{1,5,4,6,3,2\}$

Linear Interpolation



$x[n] = \{1,3,5,3,7\}$

$p[n] = \{1,0,3,0,5,0,3,0,7,0\}$ (insert zeros)

$$h[n] = \left\{\frac{1}{2}, 1, \frac{1}{2}\right\}$$

$y[n] = p[n] * h[n] = \{0.5,1,2,3,4,5,4,3,5,7,3.5\}$

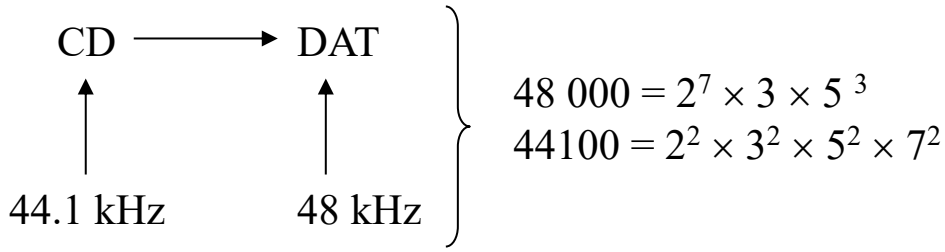
Sampling rate conversion by Non-integer factors

In some applications, the need often arises to change the sampling rate by a non-integer factor. An example is in digital audio applications where it may be necessary to transfer data from one storage system to another, where both systems employ different rates.

An example is transferring data from the compact disk system at a rate of 44.1 kHz to a digital audio Tape (DAT) at 48 kHz. This can be achieved by increasing the data rate of the CD by a factor of $\frac{48}{44.1}$, a non-integer.

In practice, such a non-integer factor is represented by a rational number that is a ratio of two integers, say L and M. The sampling frequency change is then achieved by first interpolating the data by L and then decimating by M.

Note: It is necessary that the interpolation process precede decimation otherwise the decimation process would remove some of the desired frequency components.



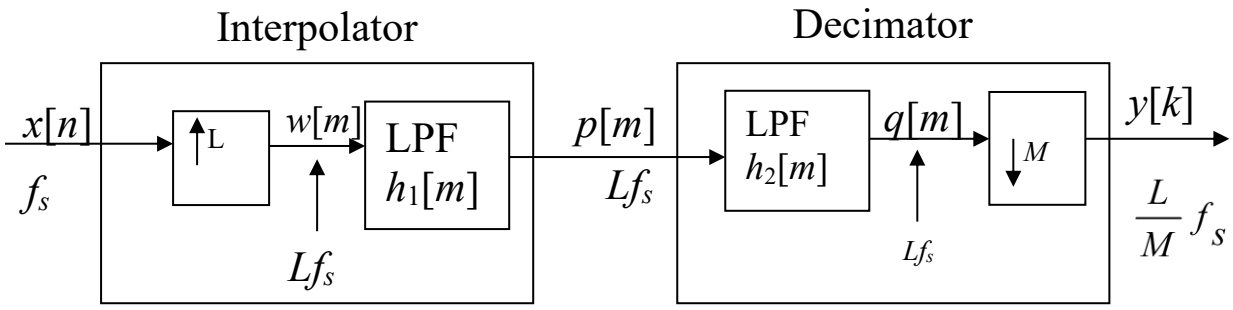
To change the sampling rate we require:

$$\frac{L}{M} = \frac{2^7 \times 3 \times 5^3}{2^2 \times 3^2 \times 5^2 \times 7^2} = \frac{160}{147}$$

Therefore if we up-sample by $L = 160$ and then down-sample by $M = 147$, we achieve the desired sampling rate conversion.

The diagram below shows that the sampling frequency change is

achieved by first interpolating the data by L and then decimating by M.

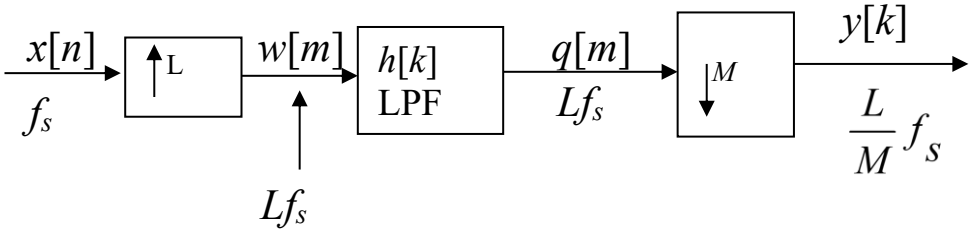


A discrete-time signal is interpolated and then decimated in order to achieve a non-integer conversion of sampling rate

The two LPFs, $h_1[m] \& h_2[m]$ can be combined into a single filter since they are in cascade and have a common sampling frequency.

If $M > L$, the resulting operation is a decimation process by a non-integer.

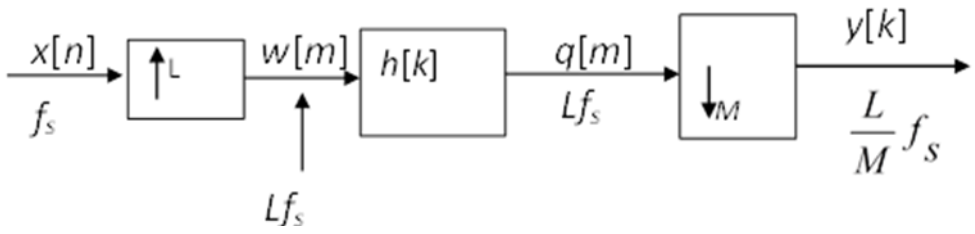
If $M < L$, it is interpolation



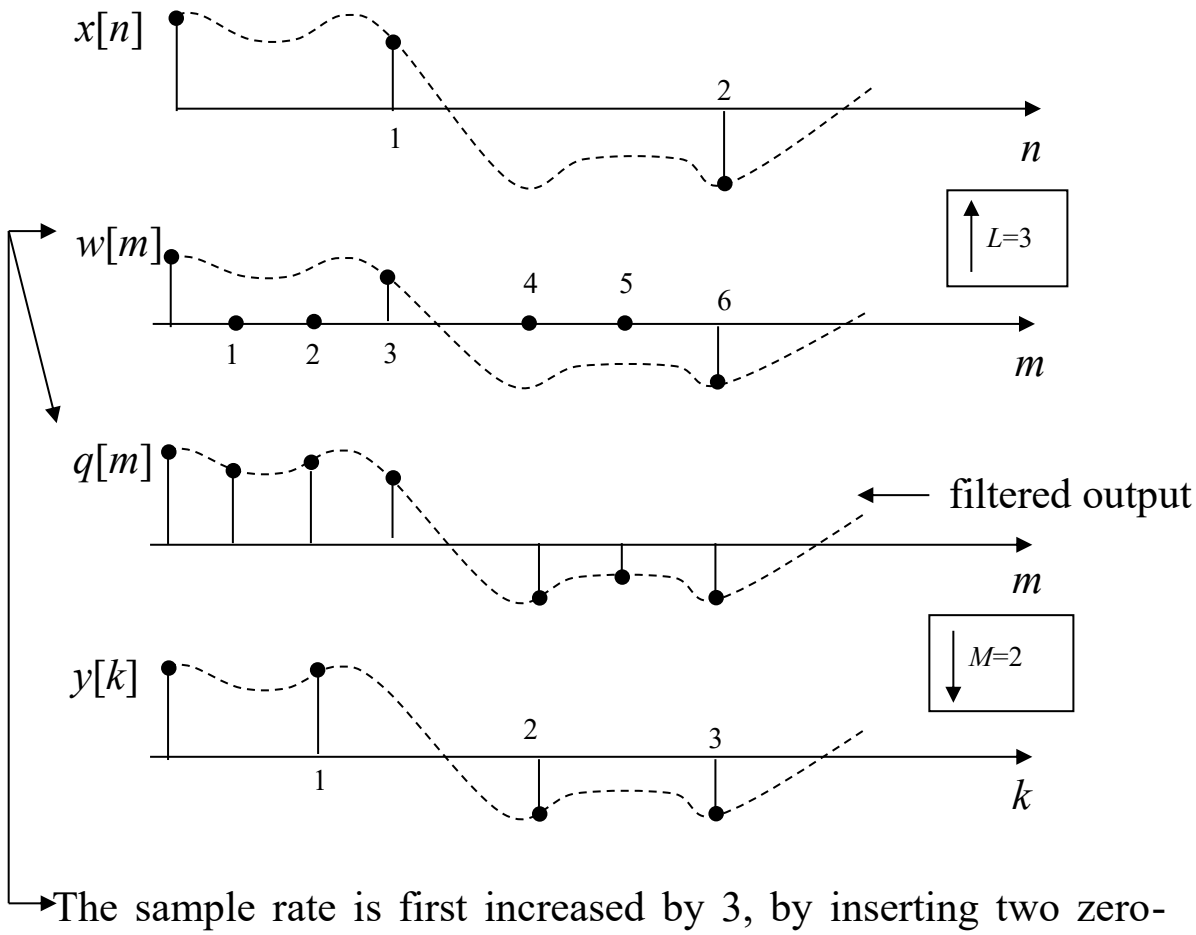
A block diagram representation of sampling rate conversion with combined low-pass filter

The lowpass filter that we require is the one that has a cut-off frequency, $\theta_c = \min\left(\frac{\pi}{L}, \frac{\pi}{M}\right)$

Illustration of Interpolation by a factor 3/2



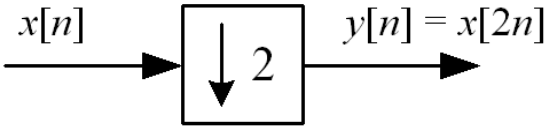
$$\frac{L}{M} = \frac{3}{2}$$



The sample rate is first increased by 3, by inserting two zero-value samples for each sample of $x[n]$ and low-pass filtered to yield $q[m]$.

The filtered data is then reduced by a factor of 2 by retaining only one sample for every two samples of $q[m]$.

Decimation by 2

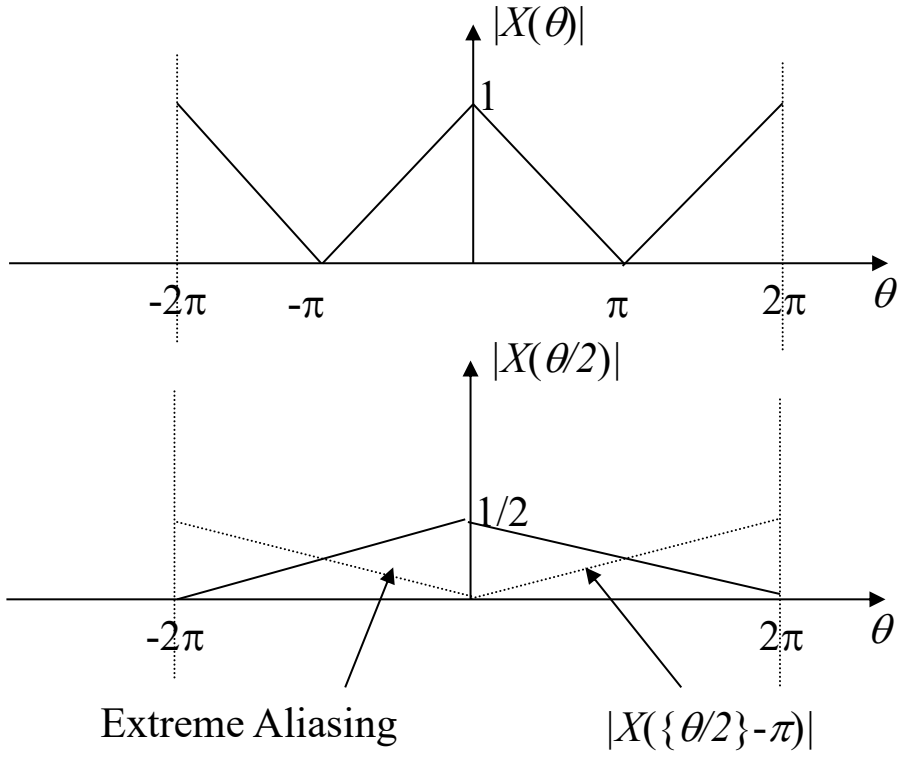


$$Y(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}}) \right]$$ Aliasing term

$$Y(\theta) = \frac{1}{2}\left[X(e^{\frac{j\theta}{2}}) + X(-e^{\frac{j\theta}{2}}) \right] = \frac{1}{2}\left[X(e^{\frac{j\theta}{2}}) + X(e^{j(\frac{\theta}{2}-\pi)}) \right]$$

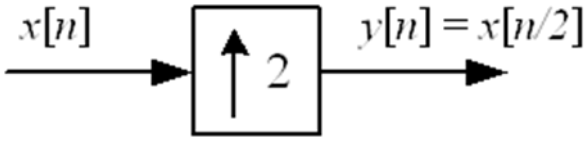
$$Y(\theta) = \frac{1}{2}\left[X(\frac{\theta}{2}) + X(\frac{\theta}{2}-\pi) \right]$$ Aliasing term

Stretch $X(\theta)$ by a factor 2 to obtain $X(\theta/2)$



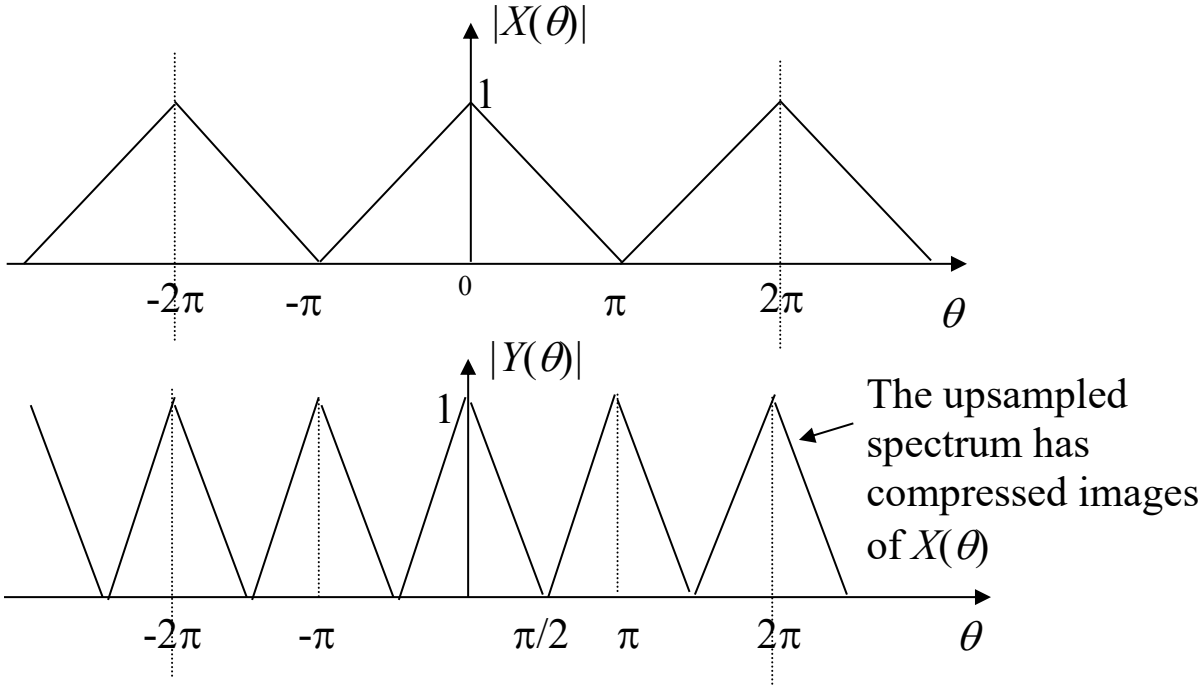
Note: The spectrum is stretched by down sampling.

Interpolation by 2



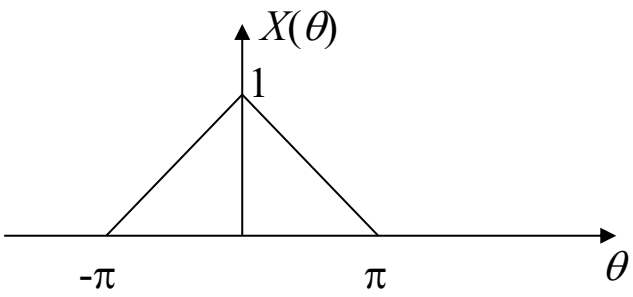
$$y[n] = x\left[\frac{n}{2}\right]; \quad Y(z) = X(z^2)$$

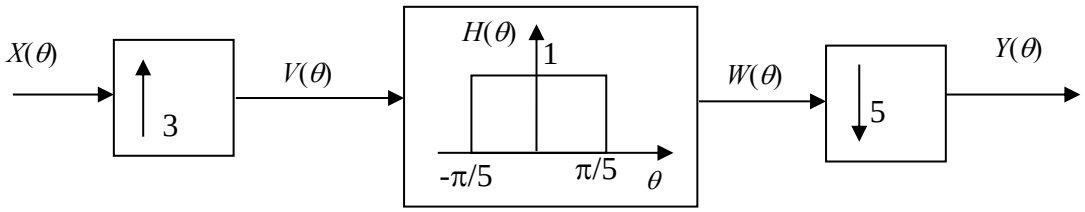
$$Y(\theta) = \left[X(e^{j2\theta}) \right] = \left[X(2\theta) \right]$$

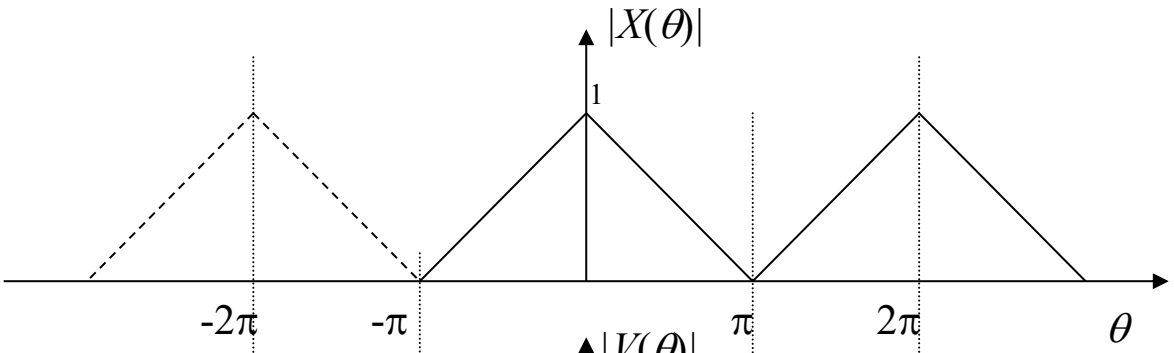


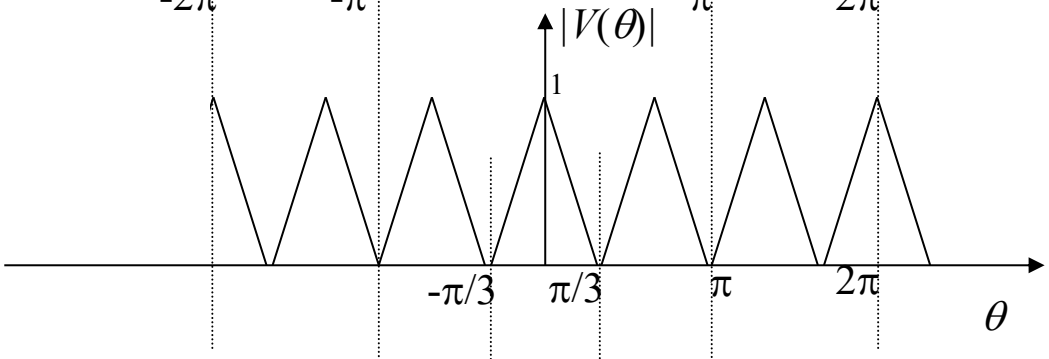
Note: The spectrum is compressed by upsampling.

<u>Example:</u> An input signal $x[n]$ with the spectrum $X(\theta)$ is applied to the system shown below. Sketch, one above another, $|X(\theta)|$, $|V(\theta)|$, $|W(\theta)|$, $|Y(\theta)|$ against θ.

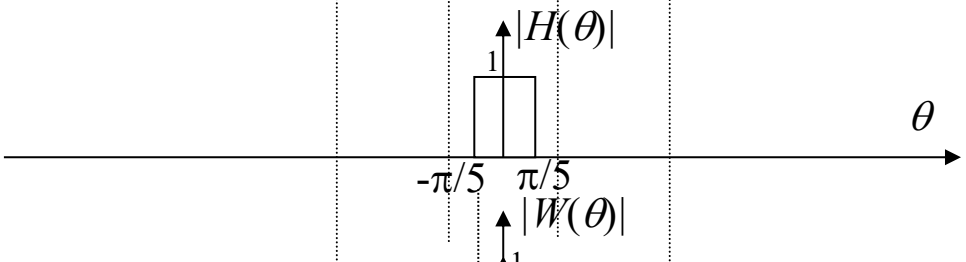


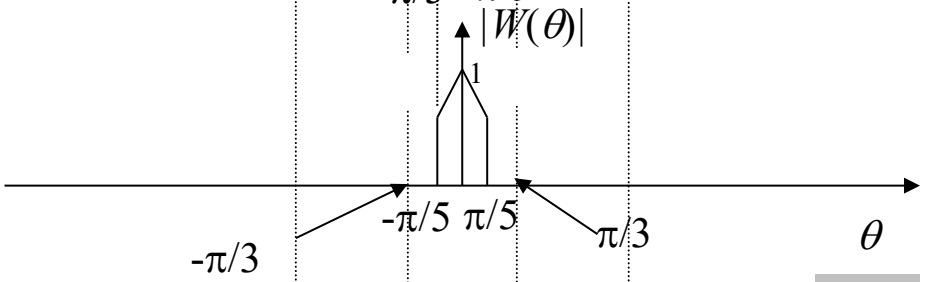

X(θ)
3
V(θ)
H(θ)
1
-π/5
π/5
θ
W(θ)
5
Y(θ)


|X(θ)|
1
-2π
-π
π
2π
θ

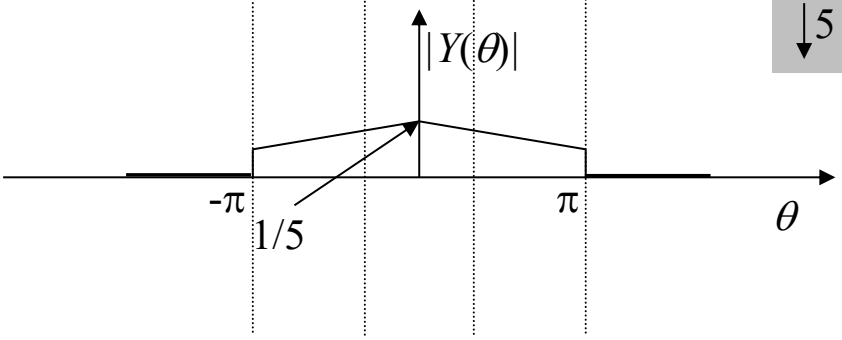

|V(θ)|
1
-π/3
π/3
π
2π
θ

↑3

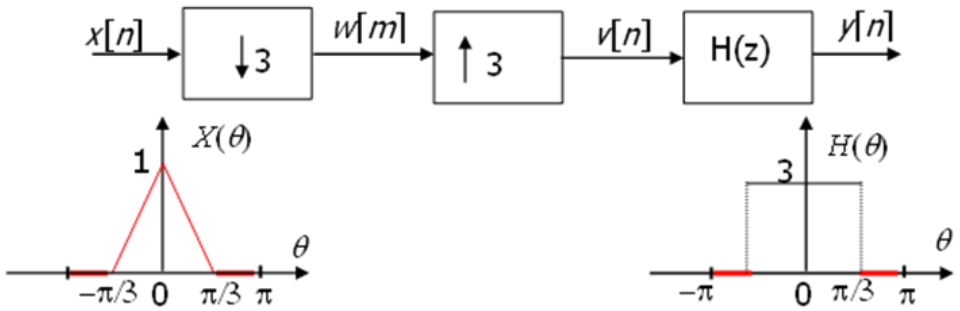

|H(θ)|
1
θ
-π/5
π/5


|W(θ)|
1
-π/5
π/5
π/3
θ
-π/3

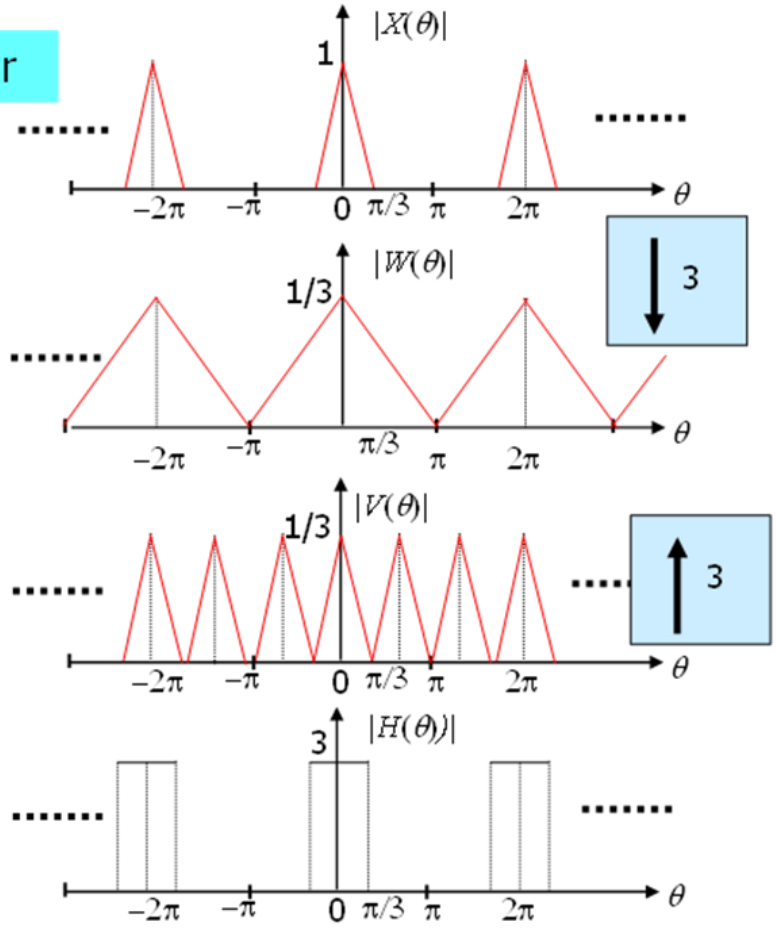
↓5

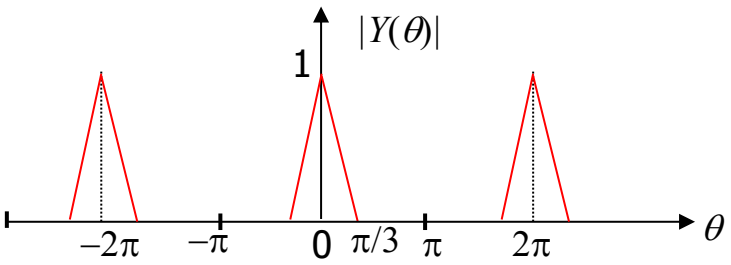

|Y(θ)|
-π
1/5
π
θ

Example: An input signal $x[n]$ with spectrum $X(\theta)$ is shown below. The input signal is applied to the system shown below. Sketch $|X(\theta)|$, $|W(\theta)|$,$|V(\theta)|$, $|Y(\theta)|$ against θ.



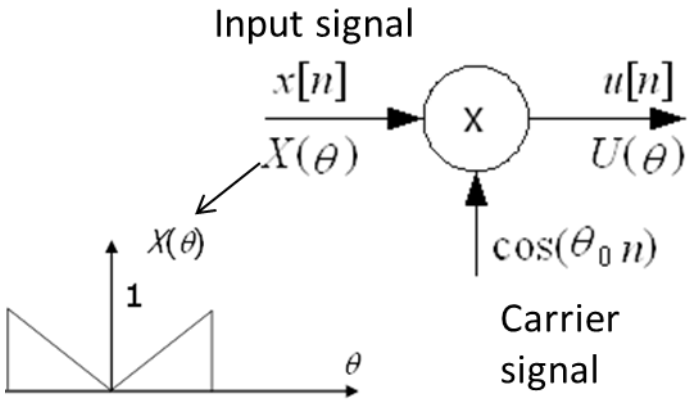
Answer





Modulation

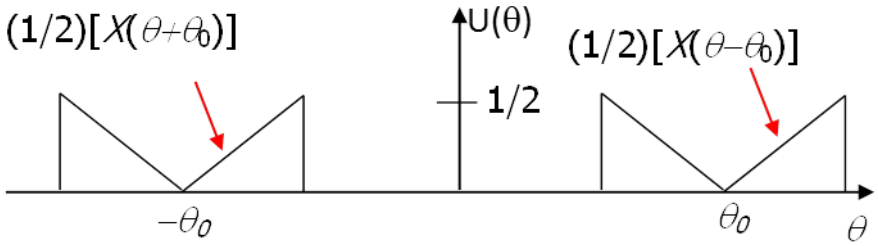
In the time domain, modulation is the process of multiplying an input signal $x[n]$ with a sinusoidal signal known as the carrier, illustrated in the diagram below.



According to the modulation property of the discrete-time Fourier transform:

$$x[n]\cos\theta_0 n \leftrightarrow \tfrac{1}{2}X(\theta+\theta_0)+\tfrac{1}{2}X(\theta-\theta_0),$$

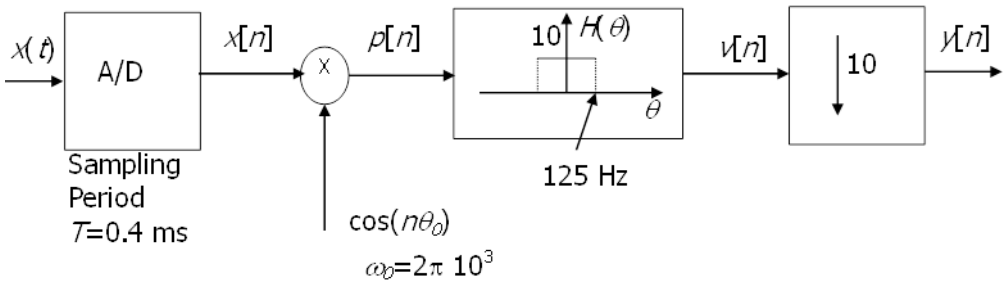
In the frequency domain, the modulated signal comprises two shifted versions of the original signal, translated by the carrier frequency.



Time Domain	Frequency Domain
$x(n)\cos(\theta_0 n)$	$\frac{1}{2}[X(\theta-\theta_0)+X(\theta+\theta_0)]$
$x(n)\sin(\theta_0 n)$	$\frac{1}{2j}[X(\theta-\theta_0)-X(\theta+\theta_0)]$
$x(n)e^{-jn\theta_0}$	$X(\theta+\theta_0)$

Example: $x(t)$ is the input signal for the system shown below. The analogue signal $x(t)$ has the spectrum $X(f)$ given by:

$$X(f)=\begin{cases}1 & 0.875kHz \le f \le 1.125kHz \\ 1 & -1.125kHz \le f \le -0.875kHz \\ 0 & \text{elsewhere}\end{cases}$$



$H(\theta)$ is an ideal lowpass filter (gain=10) with cut-off frequency f_c=125 Hz. Sketch, one above another, $|X(\theta)|$, $|P(\theta)|$,$|V(\theta)|$, $|Y(\theta)|$ against θ.

$$\text{Note}: \quad P(\theta)=\frac{1}{2}[X(\theta+\theta_0)+X(\theta-\theta_0)]$$

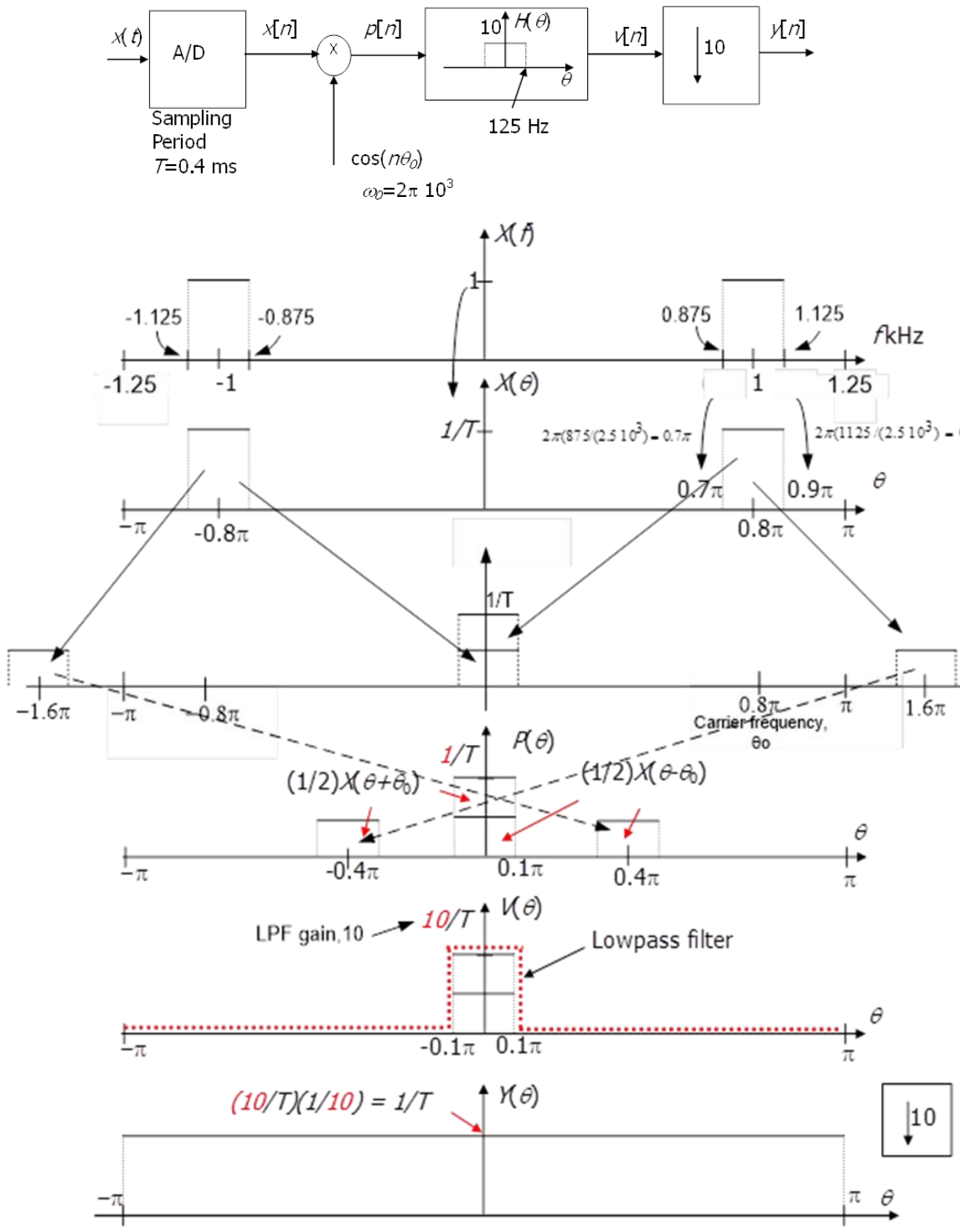
Solution:

Sampling Period, T = 0.4ms

$$f_S=\frac{1}{T}=2.5kHz;$$

Carrier Frequency: $\theta_0=2\pi 10^3(0.4\ 10^{-3})=0.8\pi;$

LPF Cutoff Frequency: $\theta_C=2\pi(125/(2.5\quad 10^3)=0.1\pi$


x(t)
A/D
x[n]
p[n]
10
H(θ)
θ
125 Hz
v[n]
10
y[n]
Sampling
Period
T=0.4 ms
cos(nθ0)
ω0=2π 10^3
X(f)
1
-1.125
-0.875
0.875
1.125
fkHz
-1.25
-1
1
1.25
X(θ)
1/T
2π(875/(2.5 10^3) = 0.7π
2π(1125/(2.5 10^3) = 0.9π
0.7π
0.9π
θ
−π
-0.8π
0.8π
π
1/T
−1.6π
−π
-0.8π
0.8π
π
1.6π
Carrier frequency, θo
P(θ)
1/T
(1/2)X(θ+θ0)
(1/2)X(θ-θ0)
θ
−π
-0.4π
0.1π
0.4π
π
V(θ)
LPF gain,10
10/T
Lowpass filter
θ
−π
-0.1π
0.1π
π
(10/T)(1/10) = 1/T
Y(θ)
10
−π
π
θ

Example

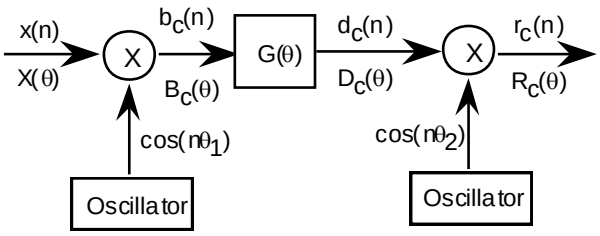
a. A signal has been sampled at 44 kHz and for a particular application this signal is required to be used at a reduced sampling rate of 8 kHz.
   (i) Draw a block diagram to show how you would achieve this.
   (ii) Indicate the type of digital filter which you would choose for part (i) above (give reasons).

b. The sampling period T of the input signal shown in the figure below is 125 μs. The relative frequency is θ = ωT. The first oscillator generates a carrier with a relative frequency $\theta_1 = \omega_1 T$, where $\omega_1 = 2\pi.2.10^3\ rad/\sec$. The second oscillator generates a carrier with a relative frequency $\theta_2 = (\omega_C + \omega_1)T$, where $\omega_C = 2\pi.10^3\ rad/\sec$. The low-pass filter has the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \dfrac{\pi}{4} \\ 0 & \dfrac{\pi}{4} < |\theta| \le \pi \end{cases}$$

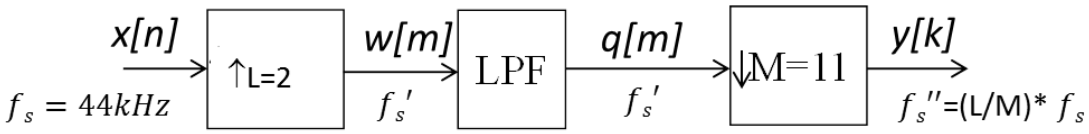
If the input spectrum is $X(\theta) = 1 + \cos(\theta) \quad -\pi \le \theta \le \pi$,

   (i) Write appropriate equations for $B_C(\theta)$, $D_C(\theta)$, and $R_C(\theta)$
   (ii) Sketch |X(θ)|, $|B_C(\theta)|$, $|D_C(\theta)|$, and $|R_C(\theta)|$ against θ,



Solution

(a)
(i)



$$\frac{L}{M} = \frac{8000}{44000} = \frac{2}{11}$$

(ii)
Using IIR filter, we cannot take advantage of the fact that we have to compute only every *N* th output, since previous outputs are needed to compute the *N* th output.

On the other hand, using FIR filter we can do our computation at the rate of fs'/N → a significantly lower computational rate.

(b)

$$\theta_1 = 2\pi \times 2 \times 10^3 \times \frac{1}{8 \times 10^3} = \frac{\pi}{2}$$

$$\theta_2 = \omega_C T + \omega_1 T = \frac{\pi}{2} + 2\pi \times 10^3 \times \frac{1}{8 \times 10^3} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$X(\theta) = 1 + \cos\theta$$

$$B_C(\theta) = \frac{1}{2}\left[X(\theta - \theta_1) + X(\theta + \theta_1)\right] = \frac{1}{2}\left[1 + \cos\left(\theta - \frac{\pi}{2}\right) + 1 + \cos\left(\theta + \frac{\pi}{2}\right)\right] = \frac{1}{2} \times 2 = 1$$

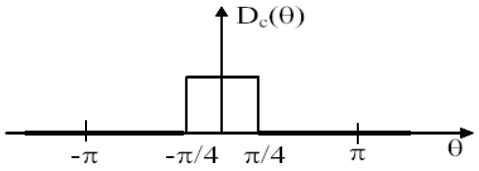
$$D_C(\theta) = B_C(\theta) \cdot G(\theta) = 1, \quad 0 < |\theta| < \tfrac{\pi}{4}$$

$$R_C(\theta)=\frac{1}{2}\left[D_C(\theta-\theta_2)+D_C(\theta+\theta_2)\right]=\frac{1}{2}\left[D_C\left(\theta-\tfrac{3}{4}\pi\right)+D_C\left(\theta+\tfrac{3}{4}\pi\right)\right]$$

$$R_C(\theta)=\begin{cases}\frac{1}{2} & -\frac{\pi}{4}\le\theta-\frac{3\pi}{4}\le\frac{\pi}{4}\Rightarrow\frac{\pi}{2}\le\theta\le\pi\\ \frac{1}{2} & -\frac{\pi}{4}\le\theta+\frac{3\pi}{4}\le\frac{\pi}{4}\Rightarrow-\pi\le\theta\le-\frac{\pi}{2}\\ 0 & \text{elsewhere}\end{cases}$$

Then from the above calculation, sketch $X(\theta), B_C(\theta), D_C(\theta),$ and $R_C(\theta)$.

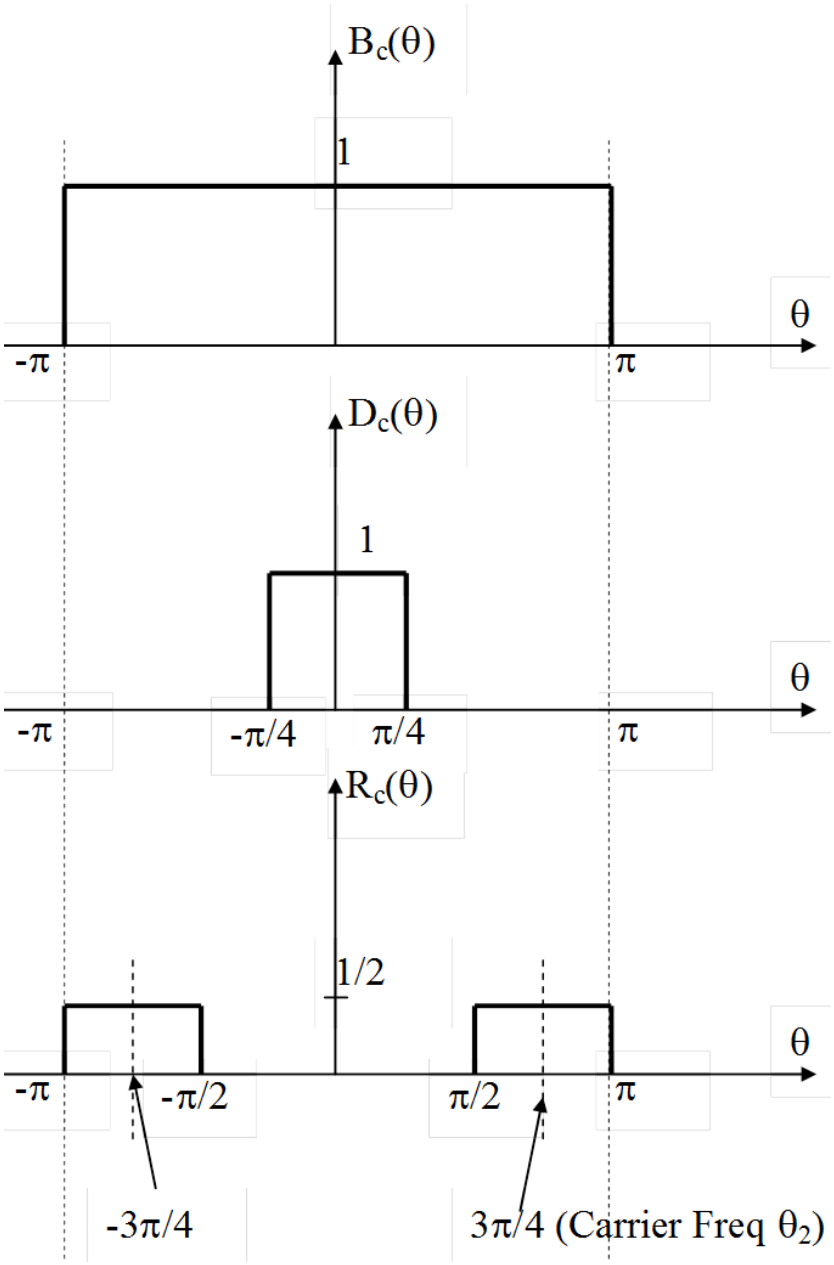
$\mathrm{B_c}(\theta)=1, \mathrm{D_c}(\theta)=1$



$$\mathrm{R_c}(\theta)=\frac{1}{2}\left\{\mathrm{D_c}(\theta-\theta_1)+\mathrm{D_c}(\theta+\theta_1)\right\}$$

$$=\frac{1}{2}\left\{\mathrm{D_C}(\theta-\frac{3\pi}{4})+\mathrm{D_c}(\theta+\frac{3\pi}{4})\right\}$$

$$R_c(\theta)=\begin{cases}\frac{1}{2} & -\frac{\pi}{4}\le(\theta-\frac{3\pi}{4})\le\frac{\pi}{4}\\ \frac{1}{2} & -\frac{\pi}{4}\le\theta+\frac{3\pi}{4}\le\frac{\pi}{4}\\ 0 & \text{elsewhere}\end{cases}=\begin{cases}\frac{1}{2} & \frac{\pi}{2}\le\theta\le\pi\\ \frac{1}{2} & -\pi\le\theta\le-\frac{\pi}{2}\\ 0 & \text{elsewhere}\end{cases}$$



End of Topic 6